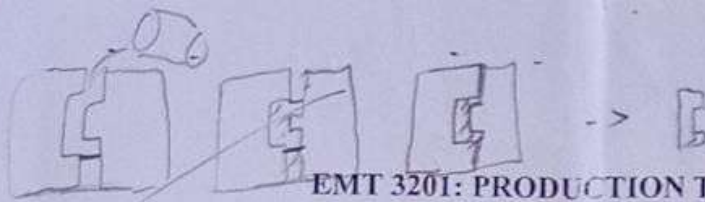


Basic permanent



EMT 3201: PRODUCTION TECHNOLOGY

Thin

low

10x10x10

- With suitable sketches, explain the following casting defects:
 - Cold shut [1 mark]
 - Misrun [1 mark]
- A perfectly plastic material is to be drawn into wire. Assuming an ideal deformation, determine the equation for the maximum possible reduction that can be achieved in one pass. [5 marks]

- Determine the value of the strain-hardening exponent for a metal that will cause the average flow stress to be $\frac{3}{4}$ of the final flow stress after deformation. [3 marks]

- A cup is to be drawn in a deep drawing operation. The height of the cup is 75 mm and its inside diameter is 100 mm. Sheet metal used in this operation has a thickness of 2 mm. If the blank diameter is 225 mm, determine:
 - drawing ratio
 - reduction
 - thickness-to-diameter ratio
 - if the operation feasible

- Derive an expression for the reduction r in deep drawing as a function of drawing ratio DR . [2 Marks]

- Outline two factors that affect the fluidity of molten metal during casting. [2 marks]

- With the aid of a neat diagram, explain the basic permanent mold casting process. [4 marks]

- Describe, with the aid of a drawing, investment casting. [5 marks]

- The total solidification times of three casting shapes made of the same casting alloy are compared: a sphere, a cylinder and a cube. The length-to-diameter ratio of the cylinder is 1.0. For all three geometries the volume is 1000cm^3 . Determine the relative solidification times for each geometry, taking the constant as 3.5min/cm^2 . [6 marks]

Handwritten calculations for the solidification times:

Volume $V = 1000\text{cm}^3$

For Sphere: $V = \frac{4}{3}\pi r^3$

For Cylinder: $V = \pi r^2 L$, where $L = d$

For Cube: $V = a^3$

Surface Area A for each shape:

Sphere: $A = 4\pi r^2$

Cylinder: $A = 2\pi r^2 + 2\pi r L$

Cube: $A = 6a^2$

Relative solidification times t_s are calculated using the formula $t_s = \frac{A}{k}$, where $k = 3.5\text{min/cm}^2$.